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Unpacking the Determinants of Financial Resilience in the Philippines

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ABSTRACT

The COVID-19 pandemic exposed financial vulnerabilities as it subjected households to health shocks and income losses. With inequalities likely to deepen, policymakers may benefit from asking: *What would make Filipinos financially resilient?* This paper examines financial resilience in the Philippines by demographic profile and employs Logistic LASSO Regression, Decision Tree, and other machine learning models to create predictive models and generate inferences on determinants of financial resilience using data from the World Bank Global Financial Inclusion (Findex) surveys for 2017 and 2021. Variables were chosen based on the components of Salignac et al. (2019)'s Multidimensional Financial Resilience Framework. Empirical findings were consistent across models and suggest that demographics may provide higher predictive value for financial resilience than financial access. Income quintile, saving behavior, and gender emerged as the top predictors in both the 2017 and 2021 survey rounds. Age, saving for retirement, and online payments were also identified as important features for 2017, and tertiary education and medical borrowing for 2021. Insights from this study could provide policymakers with baseline information on financial resilience in the Philippines and support interventions to identify and empower the financially vulnerable towards financial security.

JEL classification : D12, D14, G51, I30

Key words : financial resilience, financial vulnerability,
financial inclusion, machine learning

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I. Introduction

Posting an average economic growth of 6.4 percent (at constant 2018 prices) from 2010 to 2019, the Philippines has benefited from one of the fastest economic growth rates in the world. However, deeply entrenched inequality² persists as a conundrum to policymakers who seek to make growth more inclusive for all Filipinos (Oxford Business Group, 2020).

A World Bank (2020) survey on the impact of COVID-19 found that income losses affected 40 percent of households, especially non-farm entrepreneurs. The study further notes that “the poor and vulnerable, many of whom work in the informal sector, are especially likely to experience significant welfare losses, given their limited capacity to manage risks” (World Bank, 2020, p. 44). Inequalities are also likely to deepen in the aftermath of the COVID-19 pandemic, as pandemics have historically done (Jurzyk et al., 2020).

To increase financial safety nets for the vulnerable sector, the Philippine government has implemented various interventions even before the pandemic such as agricultural insurance subsidies (World Bank, 2020); Credit Surety Fund for micro, small, and medium enterprises (Bangko Sentral ng Pilipinas, 2018); and Emergency Cash Transfer (ECT) for targeted financial assistance to the poorest households (World Bank, 2020). Nonetheless, there have been recommendations to “assess [the] targeting of social protection programs” (Economic Policy Research Institute, 2020, p. 24) given the disparities observed between the recipients of financial assistance from different government units.

Financial inclusion, while an important contributor to financial resilience, is not sufficient to guarantee the latter. Financial inclusion refers to the accessibility, usage, and quality of financial products and services (Bangko Sentral ng Pilipinas, 2022). On the other hand, financial resilience is concerned with the ability to deal with financial shocks (Demirgüç-Kunt et al., 2022). The link between financial inclusion and financial resilience is highlighted in the Philippines’ National Strategy for Financial Inclusion 2022-2028, whose overarching vision is to steer “financial inclusion toward inclusive growth and financial resilience” (Financial Inclusion Steering Committee, 2023).

Country-level data from the 2017 World Bank Global Financial Inclusion (Global Findex) Survey³ showed that while individuals in countries with high financial access do tend to report high financial resilience, it is also possible to exhibit high access but remain low in resilience. For instance, South Africa reported 69.2 percent financial account ownership but only 28.7 percent financial resilience. The converse also holds – Vietnam was found to be low in financial access (30.8 percent account ownership) but high in financial resilience (70.0 percent).

Similarly, while the Global Findex 2021 survey finds 71 percent financial account ownership in developing economies, financial resilience in these areas stand at 55 percent (Klapper & Tayag, 2022). Half of respondents in developing

² The country’s income Gini ratio was measured at 0.45 in 1985, peaked at 0.49 in 1997, and has since gradually declined to 0.41 in 2021 (based on Philippine Statistics Authority (PSA) estimates as extracted from CEIC). The World Bank notes that the Philippines has one of the highest rates of income inequality in East Asia, second only to Thailand for countries with available data as of 2018 (World Bank, 2022b).

³ Readers may refer to Section 3 (Data) for a broad description of the World Bank Global Findex Survey.

economies were “very worried” about expenses that could arise from medical-related shocks such as illnesses and accidents (Demirgüç-Kunt et al., 2022).

As an emerging economy, the Philippine government must contend with limited capacity and resources. Policies should be crafted to provide the greatest assistance to those in greatest need. Empowering the vulnerable to achieve financial resilience is key to lifting them out of poverty and helping them stay out of it. Determining the most important environmental factors that build financial resilience and establishing the characteristics of the financially vulnerable would aid in designing better targeted interventions to curb their financial insecurity and help them achieve upward economic mobility.

This paper contributes to the literature by investigating the research question: *What makes Filipinos financially resilient?* Using 2017 and 2021 survey data from the World Bank Global Financial Inclusion (Findex) Database, this paper employed Logistic LASSO Regression, Decision Tree, and machine learning models to generate predictions and produce inferences on the most important predictors of financial resilience. This paper also contextualizes empirical results against existing financial inclusion literature, which could offer policymakers insights into the dynamics between demographics, financial inclusion, and financial resilience in the Philippines.

The rest of this paper is organized as follows. Section 2 briefly surveys related literature on financial resilience. Section 3 presents an overview of the dataset and the selection of variables used for the analysis. Section 4 characterizes the data and analyzes financial resilience by demographic profile. Section 5 explains model choice based on the data characterization in the prior section. Section 6 discusses the results of the prediction exercise and generates inferences on key explanatory variables based on the modeling results. Finally, Section 7 concludes the paper and offers potential policy implications.

II. Related Literature

Financial resilience is a multifaceted concept that has typically been characterized as the ability to withstand unexpected income loss or financial shocks (e.g., Klapper & Morduch, 2023 and Clark & Mitchell, 2022). It has also been viewed from a resource-centric perspective, focusing on individuals’ access to and usage of internal capabilities as well as external support during financial hardships (Salignac et al., 2019). Similarly, financial vulnerability has been defined as over-indebtedness, falling behind on utility payments, payments, insufficient discretionary income, and susceptibility to financial shocks (Fernandez-Lopez et al., 2023), among others.

Research on financial resilience has investigated its nexus with factors such as financial literacy, financial inclusion, financial behavior, education, income, gender, and age.

Klapper and Lusardi (2019) argued that financial literacy can strengthen financial resilience as it prevents over-indebtedness and encourages savings diversification. They found substantial gaps in financial literacy rates across income levels (i.e., between the richest 60 percent and the poorest 40 percent households) and educational attainment (i.e., between recipients of primary, secondary, and tertiary education).

Clark and Mitchell (2022) designed a financial resilience index based on questions regarding a household's ability to handle unexpected income loss, financial planning behavior, perception of debt, and overall financial concern. They found that Americans who reported higher levels of resilience at the start of the pandemic experienced lower levels of financial fragility a year after. They also found that financial fragility was lower for higher-income, higher-educated, and more financially literate households. Financial fragility is a self-reported measure of the respondents' level of confidence in their ability to gather USD2,000 in emergency funds within the next month.

Salignac et al. (2019) proposed a multidimensional financial resilience framework in their analysis of financial vulnerability in Australia. The authors developed a survey to gauge access to, or demonstration of, the framework's four components – economic resources, financial resources, financial knowledge and behavior, and social capital. The survey also accounted for the respondents' demographic characteristics, including educational attainment, home ownership, employment status, income level, gender, and age.

The study employed a linear regression model to predict composite financial resilience scores. Researchers marked responses to survey questions from 1 (severely financially vulnerable) to 4 (financially resilient), took the average score of the relevant responses for each component, and then calculated the component mean to arrive at the composite financial resilience score.

The authors found positive associations between financial resilience and both income level and education. They also found that unsatisfactory employment situations (i.e., being underemployed, unemployed, and working solely odd jobs) were linked with financial vulnerability. In addition, adults aged 18-24 obtained significantly higher resilience scores than those aged 35-49. Finally, they found no significant evidence of a gender gap.

Hussain et al. (2019) analyzed the impact of financial inclusion on financial resilience using 2014 Global Findex data for Bangladesh. The authors ran chi-square tests and logistic regression models, accounting for variables such as financial account ownership, gender, education level, income quintile, and saving behavior. They found that financial resilience rises with account ownership, education, income level, and saving behavior. They also noted a significant gender divide.

Similarly, Fernandez-Lopez et al. (2023)'s survey of financial vulnerability studies showed widespread use of (ordered) logit, (ordered) probit, and ordinary least squares regression in estimating different measures of financial vulnerability. The use of logit models was likewise noted in studies with definitions of financial vulnerability comparable to the Global Findex. Examples are Lee et al. (2019), which defined financial vulnerability as the "lack of emergency savings or rainy-day funds for three months", and Philippos & Avdoulas (2020), whose dependent variable was the self-reported inability "to raise €300 to tackle a rush next month" (as cited in Fernandez-Lopez et al., 2023).

In the Philippines, financial resilience has been identified as one of the central goals of the Philippines' National Strategy for Financial Inclusion 2022-2028 (Financial Inclusion Steering Committee, 2023). The Financial Inclusion Survey (FIS) conducted by the Bangko Sentral ng Pilipinas (BSP) has also begun capturing the concept of financial resilience in its 2021 survey round. While the topic was not covered in the nationwide quantitative survey, it was a central issue in the accompanying focus group discussions (FGDs) conducted in Mindanao in May 2022.

Apart from such efforts to understand and integrate financial resilience in policymaking, to the author's knowledge, research focusing on the topic has yet to be conducted in the Philippine context.

III. Data

Dataset Overview

Financial resilience data at the individual level remains limited in the Philippines. Nonetheless, the publicly available Global Findex Database could surface information that may be useful in assessing the determinants of financial resilience (Demirgüç-Kunt et al., 2018). The Global Findex Survey collects information on individuals' access, use, and perception of financial services and technologies. The survey also gathers data on financial resilience, which is framed as the ability of the respondent to gather emergency funds within the next month amounting to 5 percent of the country's Gross National Income (GNI) per capita. This is equivalent to around 1 month's salary for an average worker in the Philippines (Kempis & Morduch, 2020).

The Global Findex Surveys are taken from nationally representative samples of adults aged 15 and above from more than 140 economies (Demirgüç-Kunt et al., 2018). Gallup, Inc. conducts the surveys alongside the annual Gallup World Poll. Since the Findex's launch in 2011, the surveys have been completed in 3-year intervals, with the latest round delayed to 2021-2022 due to the COVID-19 pandemic⁴.

Since 2011, the global surveys have taken place through a mix of phone-based and face-to-face interviews, depending on telephone coverage and the country's conventional survey methodology. Prior to the pandemic, most developing economies were surveyed face-to-face. Several, including the Philippines, shifted to phone-based surveys in 2021 due to COVID-19 restrictions.

Data collection for the 2017 Global Findex Survey in the Philippines took place through computer-assisted face-to-face interviews of 1,000 respondents conducted from July to August 2017 (Demirgüç-Kunt et al., 2018). The surveys employed a stratified sampling approach based on population size, geography, or both. Households were chosen through random route procedures. For each household, individual respondents were randomly selected from eligible household members. The survey was carried out in seven languages, namely: Filipino, Iluko, Hiligaynon, Cebuano, Masbatenyo, Waray, and Tausug.

On the other hand, the country's 2021 survey round was conducted through phone-based interviews of 1,000 respondents in September to November 2021 (Demirgüç-Kunt et al., 2022). Sampling was done through either mobile phone random digit dialing or extraction of a nationally representative list of phone numbers. The survey was carried out in four languages, namely: Filipino, Cebuano, Bicol, and Waray.

Using the Global Findex dataset to examine the determinants of financial resilience has immediately perceptible limitations in terms of the timeliness of data collection and the narrow proxy measure for financial resilience. Moreover, model

⁴ As of writing, the World Bank has conducted Global Findex Surveys in 2011, 2014, 2017, and 2021-2022.

accuracy may be hampered by the lack of relevant demographic variables such as geographic location and occupation and predominantly binary variables.

Measurement errors may likewise arise due to self-reporting. Respondents may incorrectly recall or withhold information that they deem shameful (for instance, their ability to satisfy the condition for financial resilience). Nonetheless, the conduct of the interviews in the respondents' native languages attempts to curb misunderstanding and response bias.

Data Preprocessing

In selecting predictive attributes for the model, the primary consideration was to capture characteristics that may conceivably possess links to an individual's financial resilience. Table 1 shows the list of variables chosen for this purpose, patterned after Salignac et al. (2019)'s Multidimensional Financial Resilience Framework. These attributes are present in both the 2017 and 2021 survey datasets. Most are binary variables, except for age, which is numeric/ratio; education level, which is ordinal; and income quintile, which is ordinal.

Table 1: Predictive Attributes Selected for the Model

Economic Resources (6)	Financial Resources (4)	Financial Knowledge and Behavior (8)	Social Capital (1)	Demographic Variables (4)
• Income Quintile¹ • Received Wages* • Received Agricultural Payments* • Received Government Transfers* • Received Pension* • Mobile Phone Owner	• Financial Account Owner • Debit Cardholder • Credit Cardholder • Mobile Money Account Owner	• Saved* • Saved for Retirement* • Borrowed* • Borrowed for Medical Purposes* • Sent Domestic Remittance* • Received Domestic Remittance* • Paid Utilities* • Online Transaction*²	• Borrowed from Family/Friends*	• Age • Sex • Education Level³ • Part of Workforce

Source: Author's analysis

*denotes behavior in the past 12 months.

¹ *Income Quintile* classifies respondents into the poorest up to the richest 20 percent of households.

² *Online Transaction* is a feature generated by merging responses to the use of the internet in making bill payments and online purchases.

³ *Education level* classifies respondents into "Primary or lower", "Secondary", and "Tertiary or more". However, it is uncertain whether academic completion is required to be considered under "Secondary" or "Tertiary or more".

To eliminate missing data, observations were dropped where respondents refused to answer or did not know the answer to any of the selected attributes. This trimmed the 2017 dataset from 1,000 to 980 data points and the 2021 dataset from 1,000 to 992 data points. While both used the same 23 predictors, the target variable

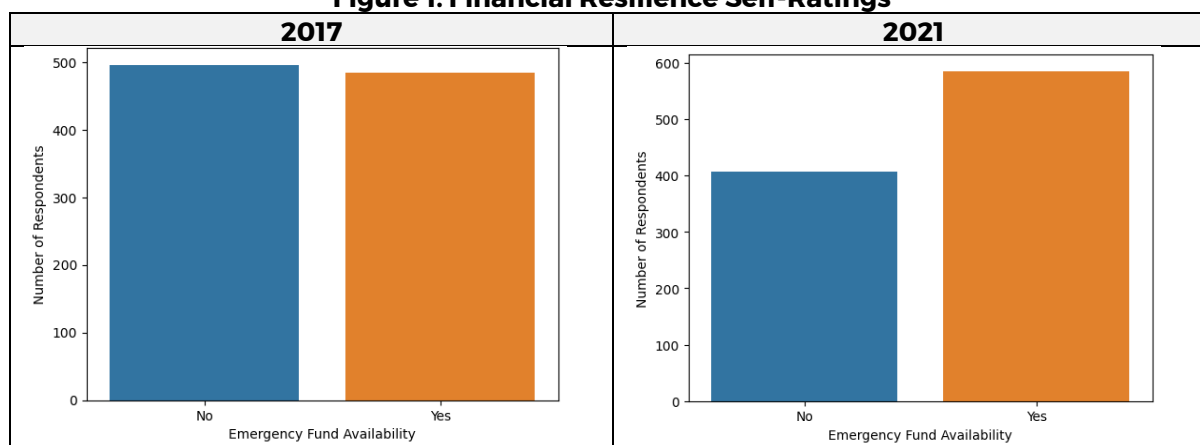
of financial resilience changed as more specific responses were introduced in the 2021 survey round⁵.

In 2017, respondents assessed their ability to gather emergency funds as “possible” or “not possible”. In 2021, the evaluation shifted to difficulty, as respondents chose between “not difficult at all”, “somewhat difficult”, and “very difficult”. For comparability across survey rounds, respondents who answered that they would encounter little to no difficulty in raising the subject funds were considered financially resilient.

IV. Descriptive Results

Financial resilience is relatively balanced in both the 2017 and 2021 surveys (Figure 1). The high frequency of both outcomes would suggest that the model may be able to learn substantially about each class to make an adequate prediction.

Figure 1: Financial Resilience Self-Ratings



Source: Author's estimates based on Global Index data

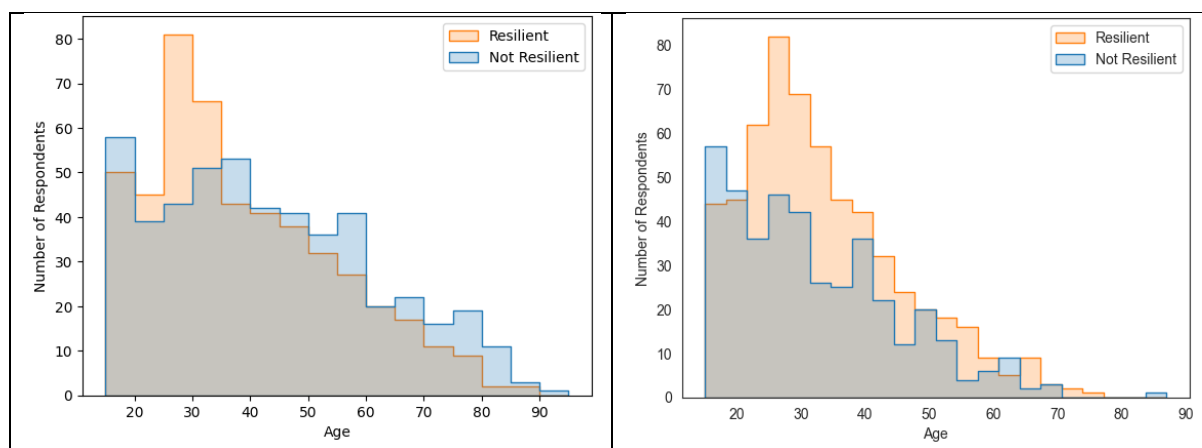
For the 2017 survey round, financial resilience is roughly equally distributed across age groups, except for respondents aged 20 to 40, who displayed a higher level of resilience (Figure 2). On the other hand, 2021 survey respondents were more financially resilient overall, except for those younger than 25 years old. Investigating the interaction between workforce participation and age in determining resilience (Figure 3), a considerably higher percentage of 2017 survey respondents who were not in the workforce reported less resilience, except for those in their late 20s to early 30s, who were markedly more financially resilient than their counterparts – possibly because their exclusion from the labor force is by choice⁶. The same general pattern holds for 2021 survey respondents.

Figure 2: Financial Resilience by Age

2017	2021
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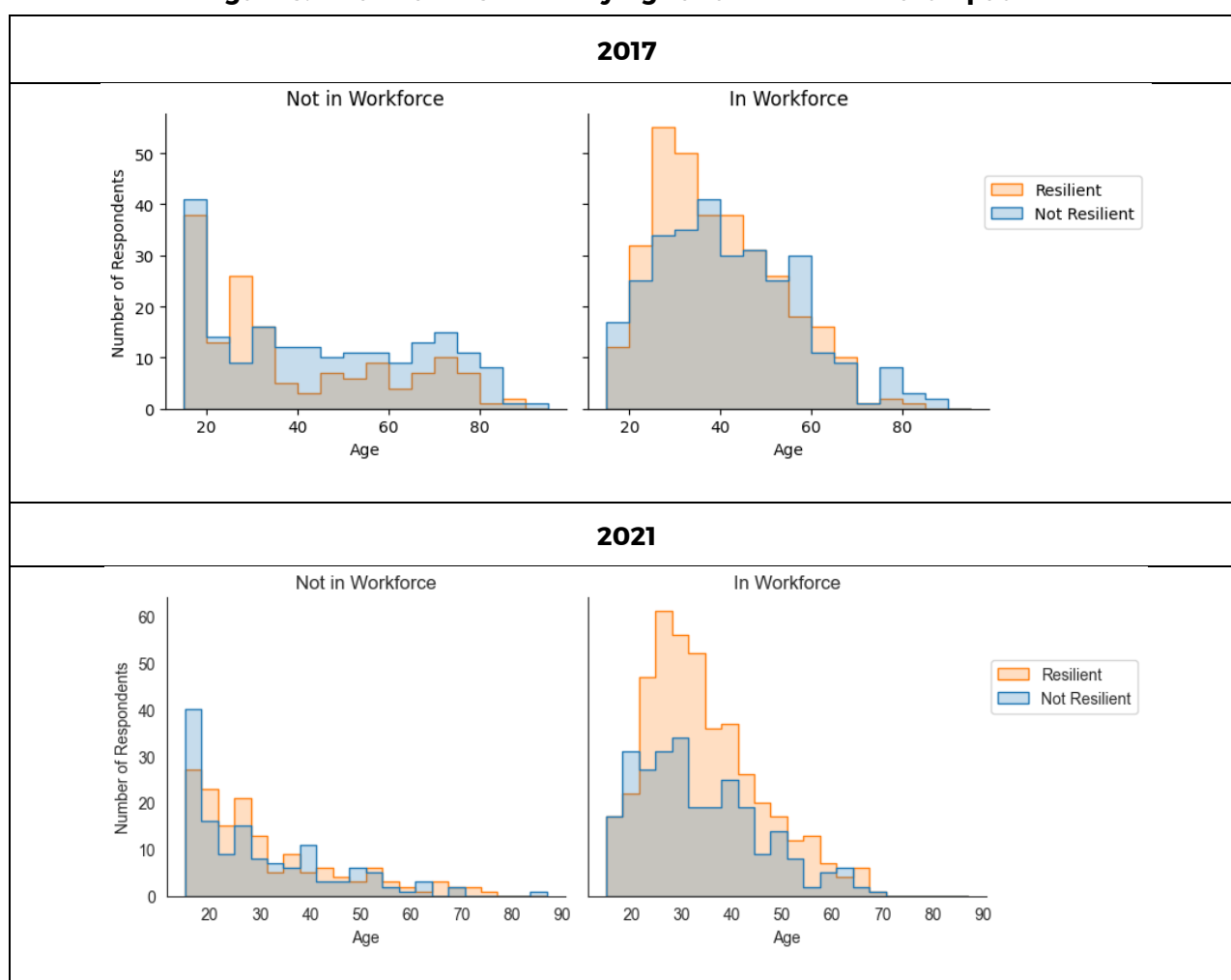
⁵ A slight modification in phrasing was also implemented in the 2021 survey round. While the 2017 survey asked about the availability of emergency funds “within the next month”, in 2021 the reference period changed to “within the next 30 days”.

⁶ The labor force comprises both the employed and the unemployed job-seekers. It excludes unpaid workers, family workers, and students. (Source: World Bank via <https://data.worldbank.org/indicator/SL.TLF.TOTL.IN>)



Source: Author's estimates based on Global Index data

Figure 3: Financial Resilience by Age and Workforce Participation

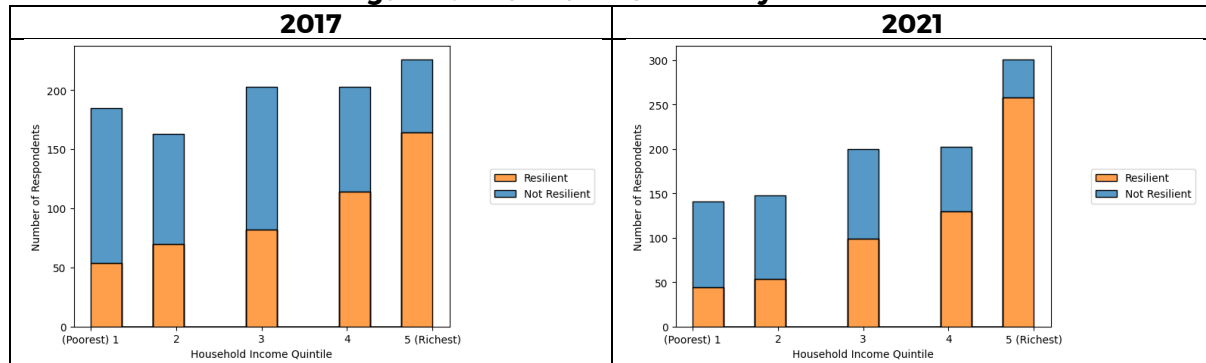


Source: Global Index, Author's estimates

A breakdown by income quintile shows a balanced dataset for the 2017 survey and a higher percentage of adults from the richest 20 percent for the 2021 survey (Figure 4). As expected, richer households reported higher financial resilience, although the proportion of non-financially resilient among the highest socioeconomic class in the 2017 survey is surprisingly elevated at 30 percent. Financial resilience also appears to have declined from 2017 to 2021 for the poorest 40 percent, and the opposite for the richest 60 percent. Examining the interaction between education and income level, the 2017 and 2021 surveys both reveal that

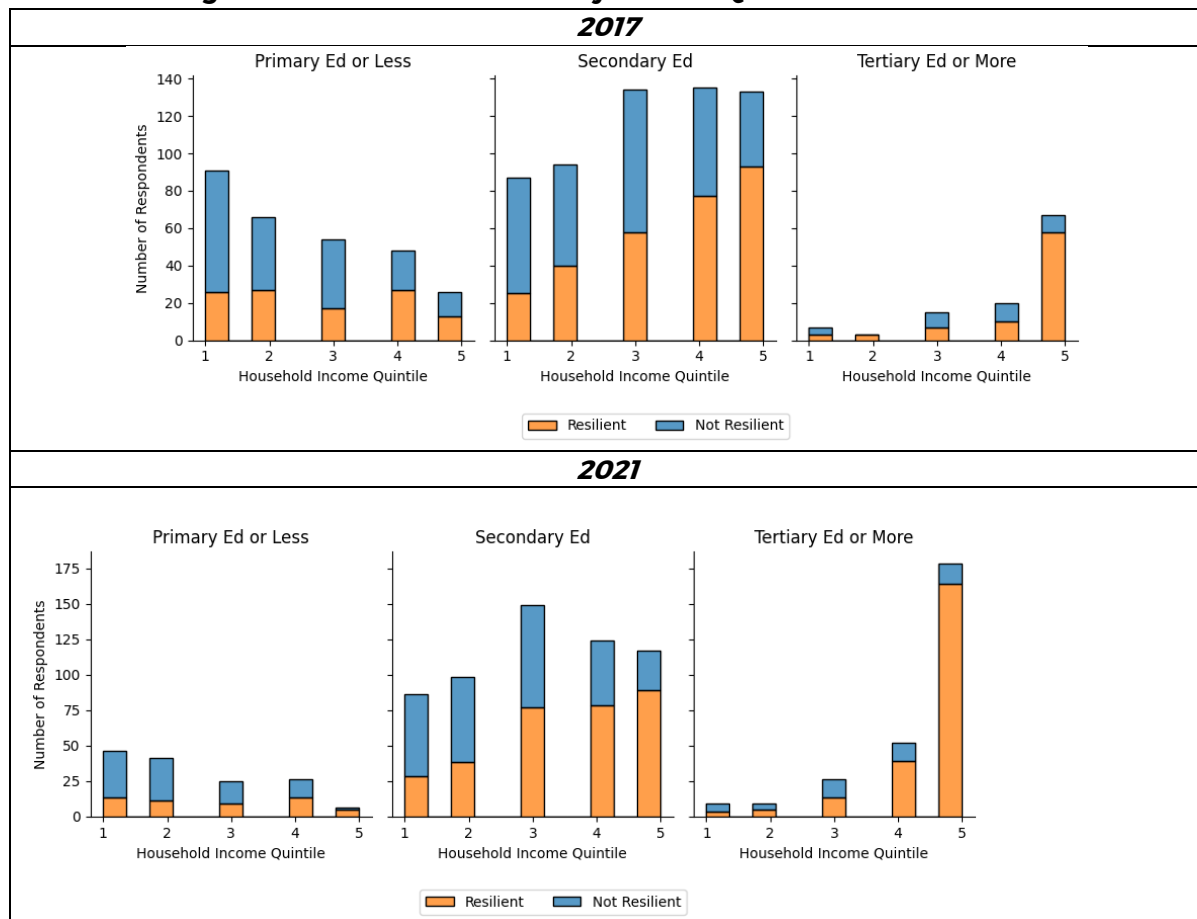
the educated tend to be more financially resilient and also belong to higher socioeconomic classes (Figure 5). These results align with Debuque-Gonzales and Corpus (2021)'s findings that lesser educated and lower-income individuals are less likely to be financially included.

Figure 4: Financial Resilience by Income



Source: Author's estimates based on Global Index data

Figure 5: Financial Resilience by Income Quintile and Education



Source: Author's estimates based on Global Index data

Gender gaps have been widely noted in much of financial inclusion literature (e.g., Sahay & Cihak, 2018). Results for the 2017 and 2021 Global Index Database show a similar gap in terms of financial resilience (Tables 2a and 2b). While the difference is less pronounced in the 2017 survey compared to other country-level studies that have used the Global Index (e.g., Hussain et al., 2019), the gender gap in financial resilience noticeably widened in 2021. These results contrast against findings from the 2017 Global Index data that females in the country are more

likely to own a bank account than males, with the gap estimated at around 10 percentage points (Debuque-Gonzales and Corpus, 2021).

Table 2a: Financial Resilience by Sex - 2017

Sex	Not Financially Resilient	Financially Resilient	Total	Financial Resilience (in percent)
Female	297	254	551	46.1
Male	199	230	429	53.6

Table 2b: Financial Resilience by Sex - 2021

Sex	Not Financially Resilient	Financially Resilient	Total	Financial Resilience (in percent)
Female	275	296	571	51.8
Male	132	289	421	68.6

Source: Author's estimates based on Global Index data

Respondents in the 2021 Global Index survey are substantially younger than those interviewed for the 2017 survey (Tables 3a and 3b). The standard deviation of 17.5 years in the 2017 survey suggests that the data is reasonably spread out and that the majority of respondents are of working age (i.e., aged 23 to 58). On the other hand, for the 2021 survey, the majority of respondents are aged 20 to 46. The younger demographic could, in part, stem from the change in interview mode due to the COVID-19 pandemic, as researchers shifted from face-to-face interviews in 2017 to phone surveys in 2021. This is consistent with reports of the Philippines having one of the highest generational divides globally both in terms of internet use and smartphone ownership.⁷

Table 3a: Descriptive Statistics for Numeric Variables - 2017

Variable	Minimum	Mean	SD	Median	Maximum
Age (in years)	15	40.5	17.5	37	95
Income Quintile	1	3.1	1.4	3	5

Table 3b: Descriptive Statistics for Numeric Variables - 2021

Variable	Minimum	Mean	SD	Median	Maximum
Age (in years)	15	32.9	12.6	31	87
Income Quintile	1	3.4	1.4	4	5

Source: Author's estimates based on Global Index data

The comparably lower median relative to mean age for both the 2017 and 2021 surveys (median of 37 and mean of 40.5 for the 2017 survey and median of 31 and mean of 32.9 for the 2021 survey) suggests that older respondents are possible outliers who are pulling the mean upwards. Indeed, both z-score and interquartile range (IQR) methods identified one respondent aged 95 as an outlier in the 2017 survey and four respondents aged 71 and above as outliers in the 2021 survey. However, these observations will be kept as they may provide useful information on one's financial resilience as one ages.

The income quintile shows a roughly uniform distribution in the 2017 survey, while the 2021 survey appears to skew towards higher income quintiles given its

⁷ A 2019 Pew Research survey found that 36 percent of Filipinos aged 50 and above used the internet or owned a smart phone, in contrast to 74 percent of 30- to 49-year-olds and 94 percent of adults aged 18 to 29 (Schumacher & Kent, 2020).

median of 4 (Tables 3a and 3b). This may likewise relate to the shift in interview mode from 2017 to 2021.⁸

Roughly 60 percent of respondents in both 2017 and 2021 surveys were classified as recipients of secondary education (Tables 4a and 4b). 2021 survey respondents tended to be more educated than their 2017 counterparts. While 29.1 percent of adults in the 2017 survey were primary-educated and 11.4 percent were tertiary-educated, the composition reversed in the 2021 survey, where 14.5 percent of respondents received only primary education or less and 27.6 percent reached tertiary education. This is in line with earlier observations of the 2021 survey demographic tending towards the technologically inclined (i.e., younger, working, and higher-income).

Table 4a: Frequency Tabulation for Education Level - 2017

Education Level	Frequency	Relative Frequency ¹ (in percent)	Cumulative Frequency	Cumulative Relative Frequency ² (in percent)
Primary or less	285	29.1	285	29.1
Secondary	583	59.5	868	88.6
Tertiary or more	112	11.4	980	100.0

Table 4b: Frequency Tabulation for Education Level - 2021

Education Level	Frequency	Relative Frequency ¹ (in percent)	Cumulative Frequency	Cumulative Relative Frequency ² (in percent)
Primary or less	144	14.5	144	14.5
Secondary	574	57.9	718	72.4
Tertiary or more	274	27.6	992	100.0

Source: Author's estimates based on Global Findex data

¹ *Relative Frequency* is the proportion of occurrence of a subject category within the dataset. It is computed by dividing the frequency of the category by the number of observations in the dataset.

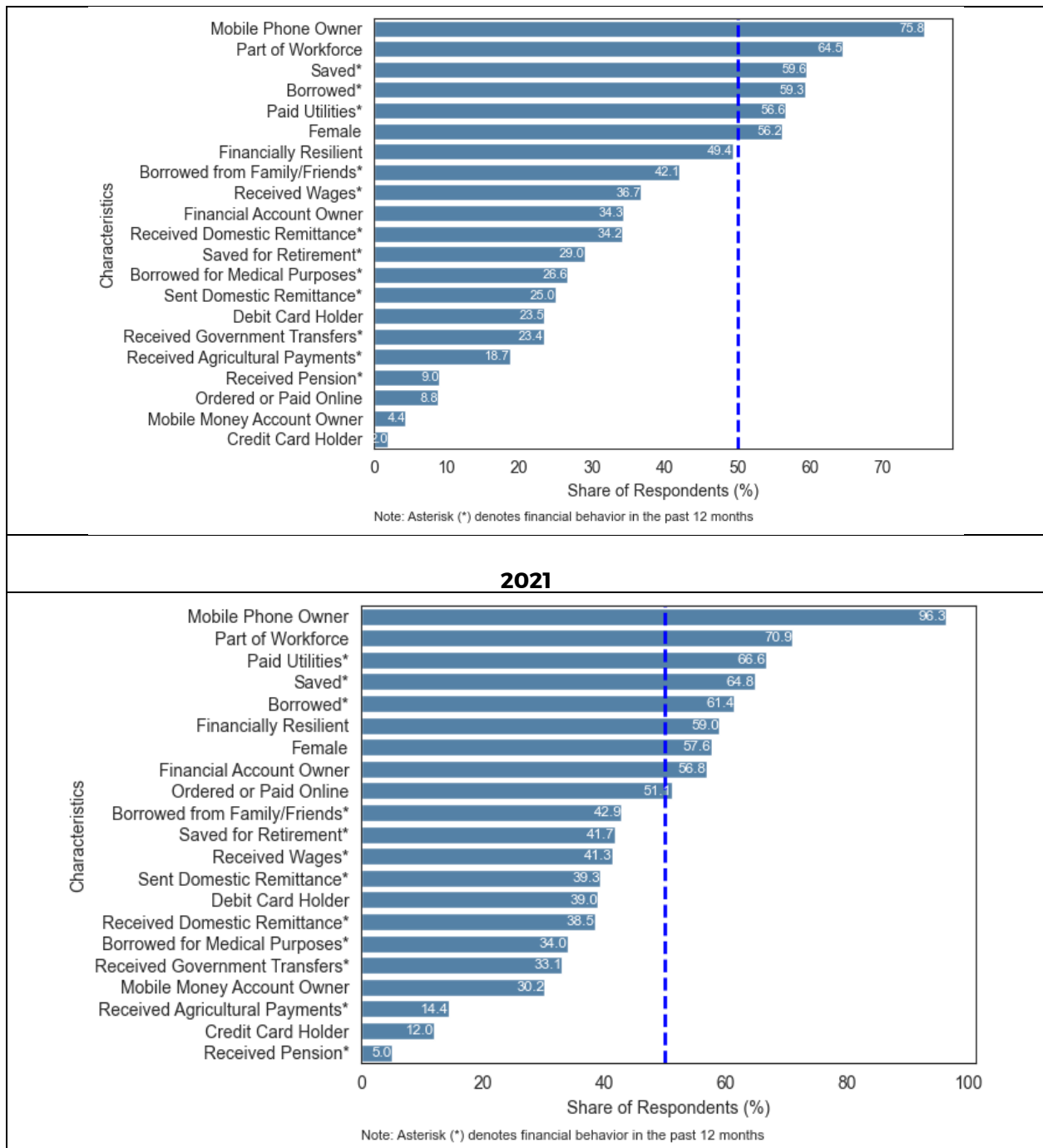
² *Cumulative Relative Frequency* is the proportion of occurrence of all categories lower than or equal to the subject category. It is computed by summing the relative frequencies of the said categories.

Figure 6 shows a graphical summary of the frequency tabulations generated by the binary variables selected for the model. More than half of indicators fall below 50 percent (marked by the blue dashed line), which highlights the long journey that remains towards financial development. This also indicates other possible rare cases that may crop up, as less than 10 percent of respondents responded positively to the bottom 4 indicators in the 2017 survey, i.e.: credit card ownership, mobile money account ownership, conduct of online transactions, and receipt of pension.

Figure 6: Profile and Financial Behavior

2017

⁸ The 2019 Pew Research survey found a differential of 19 percentage points between higher and lower-income use of internet or smartphone ownership (Schumacher & Kent, 2020).



Source: Author's estimates based on Global Findex data

Nonetheless, several indicators registered gains from 2017 to 2021 amid the COVID-19 pandemic. Mobile phone ownership rose from 75.8 percent to 96.3 percent. Financial account ownership improved from 34.3 percent to 56.8 percent. Mobile money account owners leapt from 4.4 percent to 30.2 percent. Respondents that have engaged in online purchases likewise leapt from 8.8 percent to 51.1 percent. Proactive financial behavior developed as well, with savers increasing from 59.6 percent to 64.8 percent of respondents. Saving for retirement also jumped from 29.0 percent to 41.7 percent despite the younger demographic for the 2021 survey.

These developments could, in part, be an unintended consequence of the pandemic. Social distancing protocols may have enticed people to shift to online

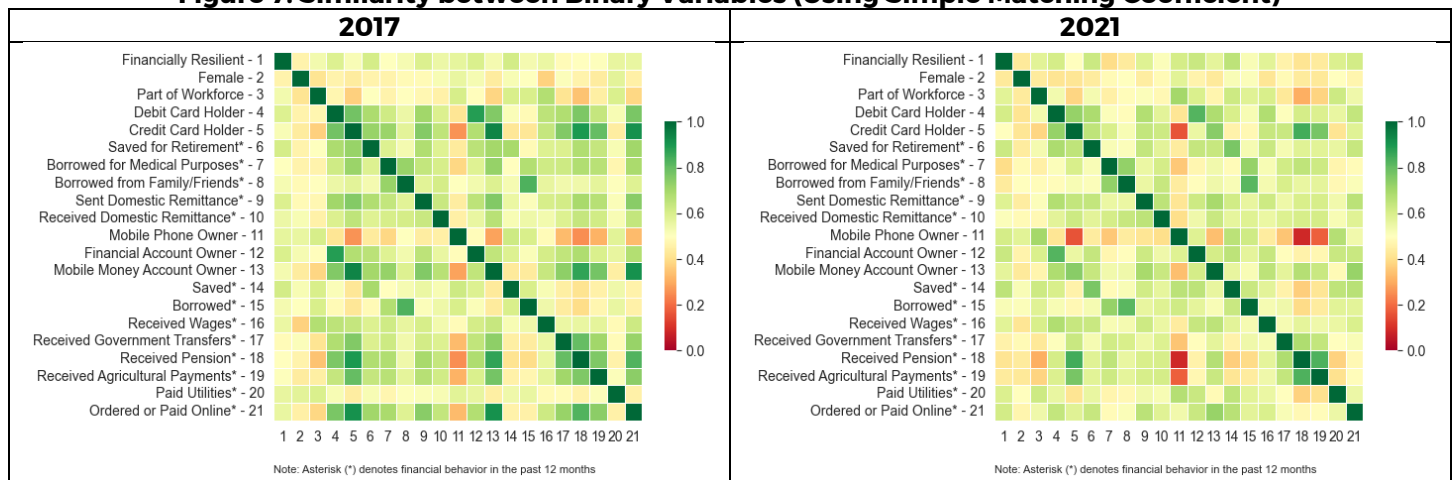
transactions, while health and income shocks may have highlighted the need for proactive saving behavior. However, the change in survey enumeration from face-to-face in 2017 to phone-based in 2021 might also have contributed to the changes. The data characterization exercise thus far consistently suggests higher technological inclination for 2021 survey respondents.

The shift to phone-based surveys may have implications on generalizability, as the 2021 respondents needed to have phone access (though not necessarily ownership) for them to be interviewed. Those without phones, likely from lower incomes or less educated backgrounds, would be excluded. Nonetheless, we could hypothesize that the income and education gaps noted in the 2021 data would widen further if those without access to mobile phones were also included in the sample.

Trends found in the Global Findex are generally consistent with the results of the BSP Financial Inclusion Survey (FIS), which found that account penetration increased from 23 percent in 2017 to 56 percent in 2021. E-money accounts likewise jumped from 8 percent in 2019 to 36 percent in 2021. However, adults with savings declined from 48 percent in 2017 to 37 percent in 2021, contrasting against the rise in saving behavior found in the Global Findex. Results of the latter suggested that savers slightly increased, from 59.6 percent in 2017 to 64.8 percent in 2021. The divergence may be due to differences in questioning, where the Findex probes into any instance of saving behavior within the past year while the FIS asks about outstanding savings at the time of the survey.

The predominantly binary nature of the Global Findex variables renders correlation (and the related Variance Inflation Factor) inappropriate in checking for multicollinearity. The Simple Matching Coefficient (SMC) would be a more suitable measure for binary attributes, as SMC calculates similarity as the proportion of exact matches between two variables (i.e., respondents whose attributes are both positive (1-1) or both null (0-0) for the variables under comparison). Based on the SMC, financial resilience does not appear to be strongly connected to any specific attribute (Figure 7). However, large SMCs (of at least 0.8) were found for combinations of least frequently occurring characteristics identified in Figure 6, which could be due to the high number of null (0-0) matches.

Figure 7: Similarity between Binary Variables (Using Simple Matching Coefficient)



Source: Author's estimates

V. Empirical Methodology

Model Selection

Based on this study's objective of deriving insights on the most important predictors of financial resilience in Filipinos, models selected for the purpose should be equipped to handle classification problems and provide meaningful inferences on the predictive power of the attributes. On the other hand, data characterization in the previous section indicates that (1) overfitting should be sufficiently addressed given the presence of 23 predictors chosen based on domain knowledge; (2) the model would ideally be robust to outliers, considering rare binary cases and a relatively small percentage of older respondents; (3) the model should be able to handle redundant/irrelevant variables, given the various related financial indicators selected as attributes; and (4) the model should be capable of dealing with interacting variables, whose presence is evident from the data characterization. Thus, the Logistic/Logistic LASSO Regression and the Decision Tree models appear well-suited for this study.

Logistic and Logistic LASSO Regression

Logistic regression belongs to a set of parametric techniques widely used as a foundation for statistical analysis due to their interpretability and portability from theoretical to empirical analysis.

Logistic regression predicts the probability that a particular entity belongs to a specific class. It does so by generating maximum likelihood estimates based on the combination of characteristics that led to previous occurrences of the outcome. In other words, it determines likely predictors of known outcomes and assesses new data against this model to predict which outcome is most likely given a specific set of values for the predictors.

Logistic regression has also been used in similar studies analyzing financial inclusion. One such study uses Findex Data collected from selected Southeast Asian countries in 2014 to study the determinants of financial inclusion, defined as access to a financial account (Tjahjadi & Ajani, 2018). The research used individual characteristics and factors related to borrowing to predict the level of inclusion across different income levels and countries.

In this study, logistic regression was chosen because it deals well with irrelevant, redundant, and interacting variables. However, the technique tends to be sensitive to outliers. Hence, issues may arise around the robustness of the results amidst the presence of rare cases in the age variable as well as some financial indicators.

Since it seeks the combination of predictors that will best fit existing data, logistic regression models tend to overfit as the parameters increase. To aid in model generation, LASSO can be used as a shrinkage method to zero out irrelevant coefficient estimates. LASSO can filter factors of financial resilience that provide the highest predictive value. Performing LASSO requires standardized features to eliminate issues with scaling that may cause larger-ranged variables such as age to dominate the regression model.

Decision Tree

Decision tree is a non-parametric technique that splits known cases into groupings that would maximize homogeneity. New data will then be classified based on rules generated from the splits. Decision tree is widely used for its straightforward interpretability and accessibility to the non-technical audience. The method is likewise robust to irrelevant variables and outliers, which is important in this study given the relatively large number of parameters and the rare cases noted during data characterization.

Decision tree has likewise been used to determine key factors in financial inclusion. Aiming to segment Indian survey respondents into the financially excluded and included, Tiwari et al. (2019) generated classes based on a composite measure of financial inclusion (composed of bank account ownership, access to loan requirements, insurance coverage, and access to digital banking). The authors generated a decision tree that classified individuals as included or excluded based on demographic, social, and economic variables.

Like logistic regression, decision tree may be prone to overfitting as the number of parameters increases. Too many variables may cause the decision tree to accidentally overstate the importance of irrelevant variables. Nonetheless, this should not preclude the use of the algorithm. Various methods can address model overfitting, such as limiting tree size or pruning to penalize larger trees.

VI. Results and Discussion

Predictive Models of Financial Resilience

Model performance is evaluated against the baseline financial resilience rate (49.4 percent in the 2017 survey and 59.0 percent in the 2021 survey). This means that if a model were to tag all respondents as financially resilient, it will obtain an accuracy of 49.4 percent on 2017 data and 59.0 percent on 2021 data. Similarly, if all respondents are labeled as financially vulnerable, the accuracy will stand at 50.6 percent on 2017 data and 41.0 percent on 2021 data.

For model building, this study runs different logistic regression and decision tree models based on selected hyper-parameters. Hyper-parameter values maximizing test set accuracy were chosen. The chosen models were then run against the same set of training and test data, split at 25 percent to allow the models to train on a substantial training set (more than 700 observations) while keeping a sizeable test set (almost 250 observations).

Tables 5a and 5b compare the accuracy scores of the different models on the test data. The accuracy scores hovered in the 59-65 percent range for 2017 data and 62-68 percent for 2021 data. To the author's knowledge, the closest study on financial resilience that similarly gauges model accuracy is that of Hussain et al. (2019), which estimates that its logistic model can correctly identify the financial resilience of 76 percent of individuals, based on 2014 Global Findex data in Bangladesh.

Table 5a: Accuracy Scores across Different Models - 2017

Classifier	Hyper-parameter ¹	Accuracy Score (in percent)
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Logistic Regression	<i>none</i>	60.0
	LASSO (L1 penalty) with C = 0.21	61.2
Decision Tree	Maximum Tree Depth = 2	61.2
	Minimum Samples per Leaf = 39	63.7
	Cost Complexity Pruning Alpha = 0.015	61.2
	Minimum Samples per Split = 151	64.1
	Grid-Search Optimized Parameters	59.2
Random Forest ²	Maximum Tree Depth = 3; Estimators = 1,500	65.3

Table 5b: Accuracy Scores across Different Models – 2021

Classifier	Hyper-parameter ¹	Accuracy Score (in percent)
Logistic Regression	<i>none</i>	66.5
	LASSO (L1 penalty) with C = 0.1	66.5
Decision Tree	Maximum Tree Depth = 4	65.3
	Minimum Samples per Leaf = 52	66.9
	Cost Complexity Pruning Alpha = 0.015	65.3
	Minimum Samples per Split = 99	68.1
	Alpha = 0.001; Minimum Samples per Leaf = 60	66.9
Linear Discriminant Analysis ³	<i>none</i>	66.9

Source: Author's estimates

Table 5 Notes:

¹ Except for Logistic Regression, models were iterated across a range of hyper-parameter values. Final hyper-parameter values were those that yielded the best mean test set accuracy scores.² Random Forest generates predictions based on an ensemble of Decision Trees trained on different subsets of the data and variables.³ Linear Discriminant Analysis classifies data based on linear combinations of variables that minimize within-group separation and maximize between-group separation.

Since both decision tree and logistic regression models may sacrifice accuracy for interpretability, a benchmark model was introduced to determine whether higher predictive accuracy could be achieved. Random Forest (with a maximum tree depth of 3 and 1,500 estimators) emerged with the best predictive score on 2017 data and Linear Discriminant Analysis on 2021 data. Nonetheless, both models still resulted in test accuracy scores below 70 percent (Table 5). Other models considered include Naïve Bayes, Linear Discriminant Analysis, K-Nearest Neighbors, Neural Networks, Random Forest, and Support Vector Machines⁹.

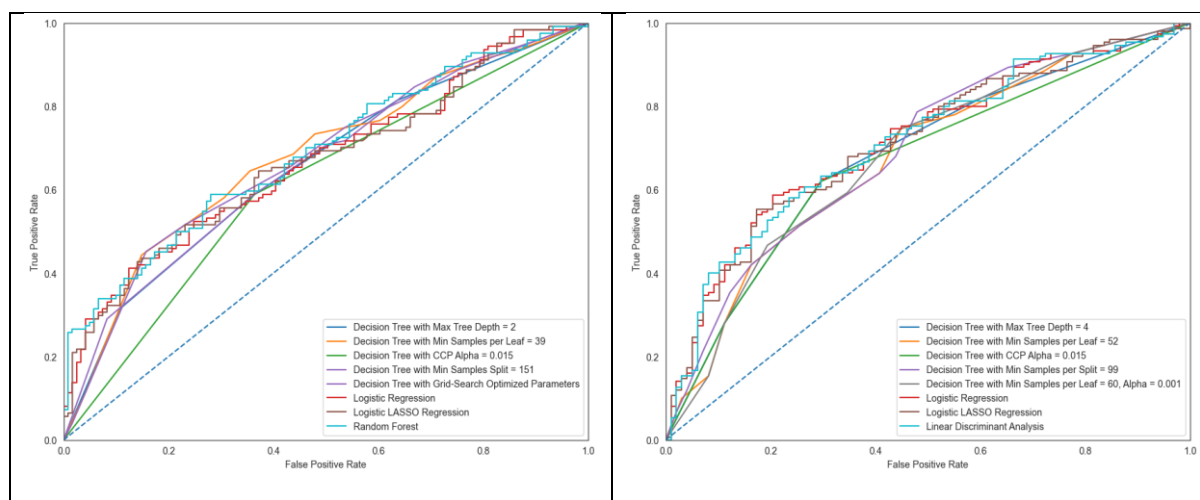
The Receiver Operating Characteristic (ROC) curve in Figure 8 provides a graphical comparison of the models' predictive performances. All models follow roughly the same path – all better than 50 percent chance but far from the ideal where the curve hugs the top left corner of the axis (signifying the ability to achieve high true positive rates while keeping false positive rates low). The Random Forest tends to produce better results for 2017 data while Linear Discriminant Analysis and Logistic (LASSO) Regression tend to perform better for 2021 data.

Figure 8: Receiver Operating Characteristic (ROC) Curve

2017	2021
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⁹ Accuracy scores for the other models are as follows:

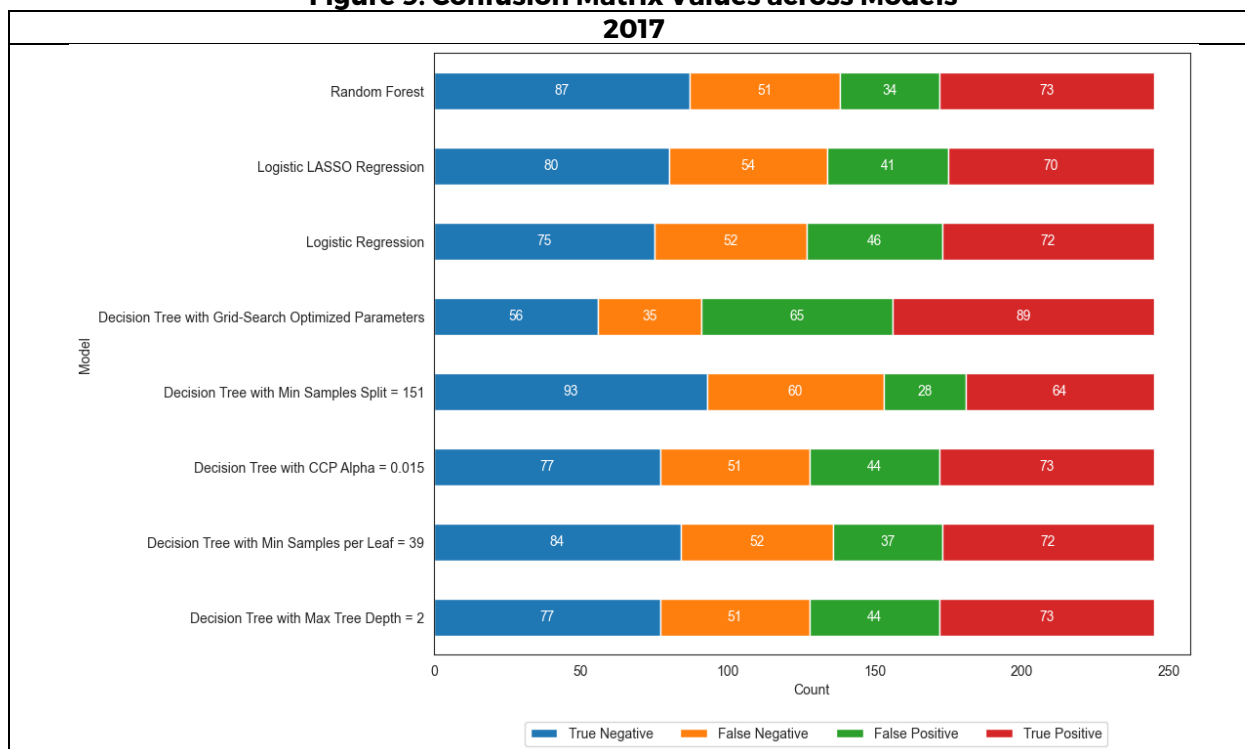
- For 2017 survey data, Linear Discriminant Analysis: 60.0 percent, K Nearest Neighbors: 62.9 percent, Support Vector Machine: 60.0 percent, Naïve Bayes: 62.0 percent, Neural Network: 63.7 percent
- For 2021 survey data, K Nearest Neighbors: 65.7 percent, Random Forest: 64.1 percent, Support Vector Machine: 65.7 percent, Naïve Bayes: 65.3 percent, Neural Network: 64.9 percent

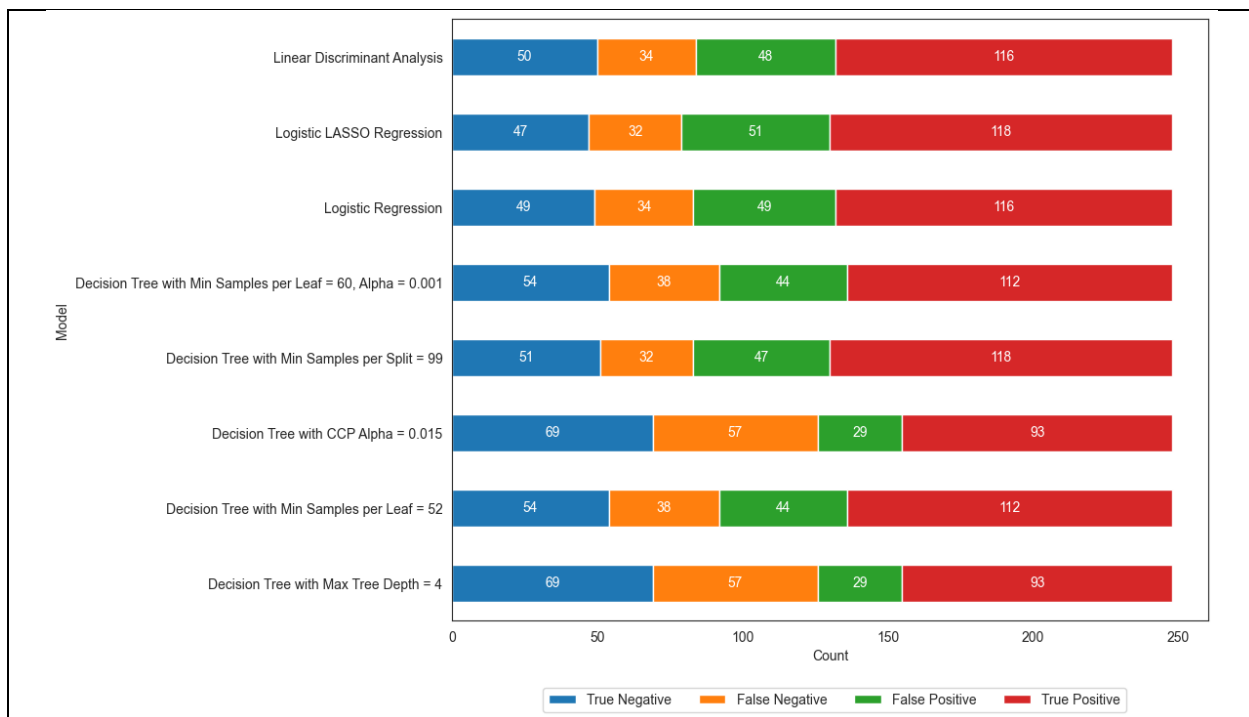


Source: Global Findex, Author's estimates

While accuracy is important in distinguishing the financially resilient from the financially vulnerable (which would thereby facilitate the proper targeting of interventions), it is more important to dissect the components of these accuracy measures and assess the nature of the models' misclassifications. The models' confusion matrices highlight the trade-off between sensitivity (ability to identify true positives) and specificity (ability to identify true negatives) (Figure 9). For 2017 data, the Decision Tree model with at least 151 samples per split yielded the highest number of true negatives and fewest false positives (where "positive" means financially resilient and "negative" means otherwise). For 2021 data, the same is exhibited by the Decision Tree model with cost complexity alpha of 0.015.

Figure 9: Confusion Matrix Values across Models





Source: Author's estimates

Taken together, the model performance metrics considered thus far indicate that the best overall model for the 2017 data would be the Decision Tree model with at least 151 samples per split as it achieved the highest accuracy score while maximizing true negatives and minimizing false positives. The ROC curve likewise showed that, except at very low positive rates, the same Decision Tree model generally performed at par with the Random Forest, which registered the best performance in this measure. On the other hand, for 2021 data, considered model metrics suggest that the best overall model would be the Decision Tree model with at least 99 samples per split. It topped accuracy at 68.1 percent and performed well at higher positive rates in the ROC curve. It also identified the highest number of true positives out of the models tested, but fewer true negatives than the Decision Tree model with cost complexity alpha of 0.015.

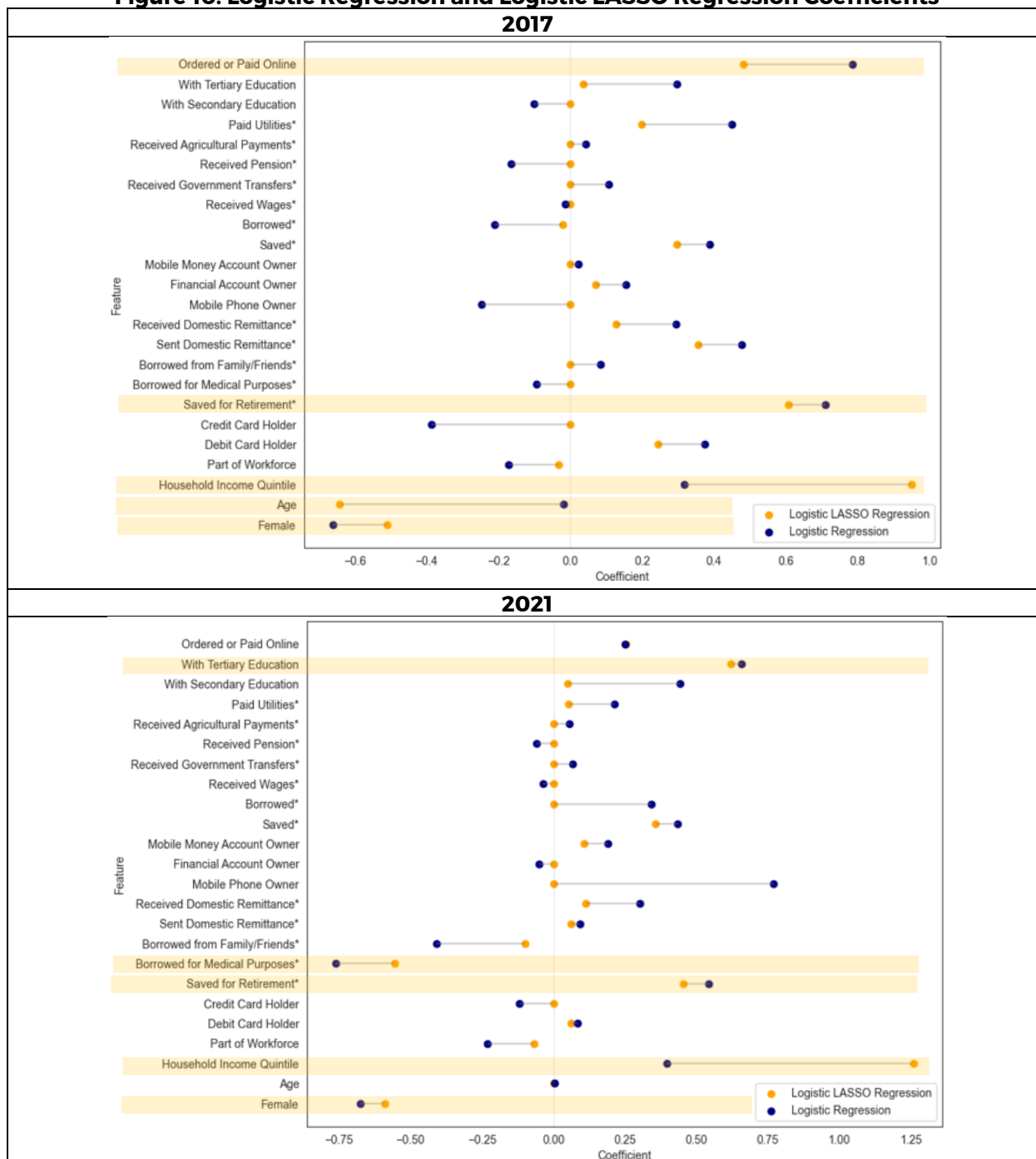
Nonetheless, with the intention is to cover as many of the financially vulnerable as possible, the models chosen to generate inferences in the next section were those that identified the highest number of true negatives and lowest false positives, i.e.: for 2017, the Decision Tree with at least 151 samples per split (which incidentally is also the best overall model) and for 2021, the Decision Tree with cost complexity alpha of 0.015. It may be worth noting that the tradeoff for choosing a model with a higher number of false negatives would be that interventions would also reach a considerable number of individuals who may already be financially resilient.

Inferences on the Determinants of Financial Resilience

Logistic LASSO regressions for both the 2017 and 2021 survey rounds consistently suggest that financial resilience possesses a strong, positive relationship with income quintile; a moderate, positive relationship with saving for retirement; and a moderate, negative relationship with being female (Figure 10). The magnitude of the regression coefficient for income quintile noticeably rose from 2017 to 2021. Conversely, the magnitude for saving for retirement decreased

and for gender (female), increased slightly. Moreover, tertiary education jumped from assuming a minimal regression coefficient in 2017 to being second only to income quintile in 2021. The widening financial resilience gap between income classes, genders, and educational attainment, coupled with the weakened link between financial resilience and saving behavior, supports concerns that the COVID-19 pandemic would likely deepen inequalities (Jurzyk et al., 2020).

Figure 10: Logistic Regression and Logistic LASSO Regression Coefficients



Source: Author's estimates

The LASSO regression found that age was strongly negatively associated with financial resilience in 2017 but turned uninformative in 2021. In contrast, the model found borrowing for medical purposes uninformative in 2017 but noted a stronger negative association with financial resilience in 2021, second only to gender (i.e.,

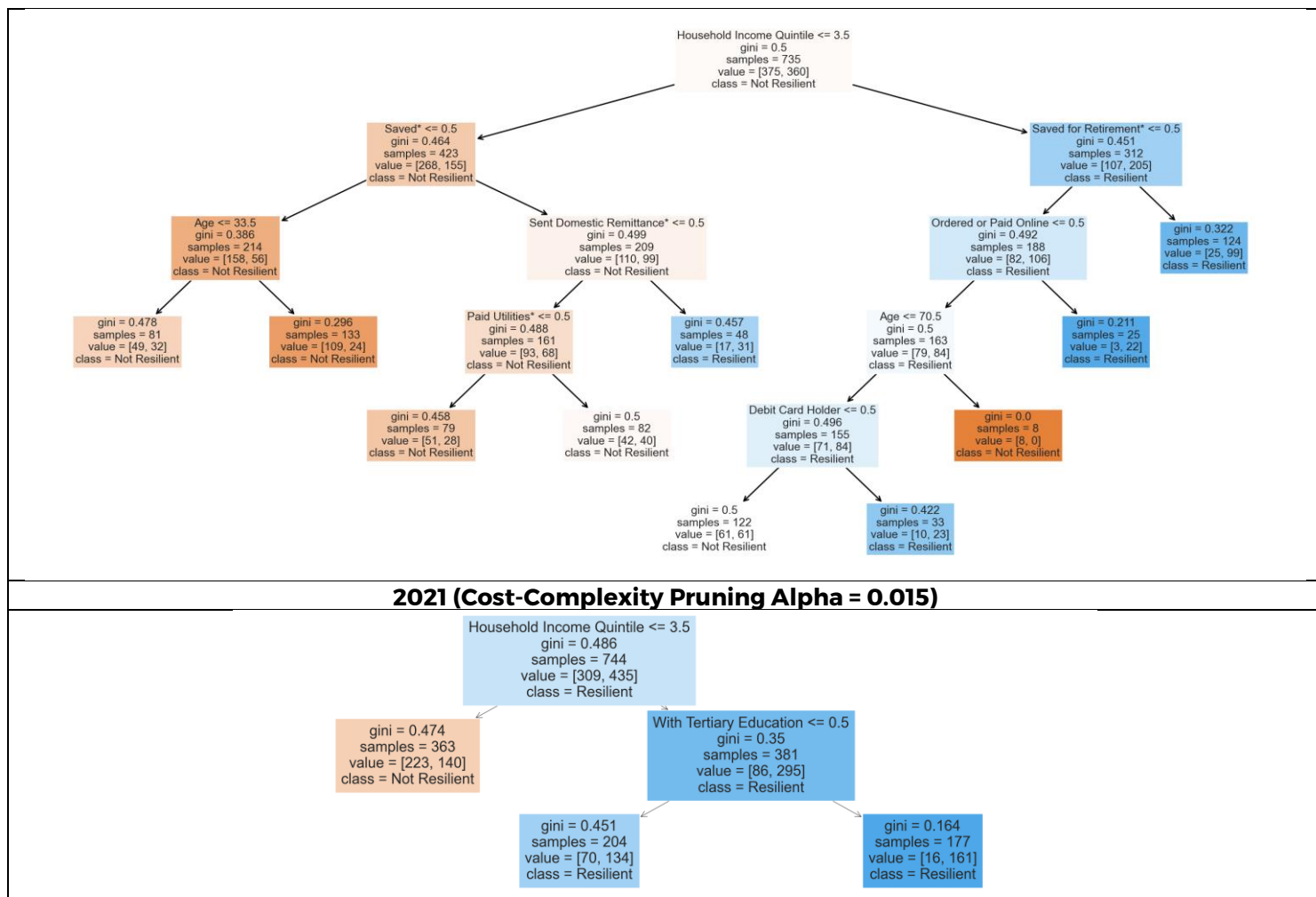
being female). The change in predictive value for age might be linked with the earlier result where respondents in the 2017 survey aged below 40 tended to be more financially resilient than older age groups, especially those in retirement age (70 and above). On the other hand, there were no discernible differences between age groups in the 2021 survey. The negative association found for medical-related borrowing in the 2021 survey could be linked with findings of the COVID-19 Household Survey that the World Bank conducted in December 2020, where 62 percent of households cited lack of money as a reason for inability to access treatment (Piza et al., 2021).

While online payments was one of the top predictors in the 2017 survey, its association with financial resilience decreased in 2021. This may be due to the adjustments brought about by physical lockdowns during the COVID-19 pandemic, as transacting online became widespread. The Global Findex survey found that Filipinos who conducted online payments sharply increased from 9 percent in 2017 to 51 percent in 2021. That the pandemic transformed online transactions into a norm rather than an exception may have led to the financial indicator's weakened ability to distinguish between the resilient and vulnerable (perhaps as a proxy for financial literacy).

The positive association between financial resilience and sending domestic remittances also noticeably decreased from 2017 to 2021. Like online payments, the Global Findex found that more Filipinos sent domestic remittances during the pandemic, from 25 percent in 2017 to 39 percent in 2021. While the reason for the remittances is unknown, its positive link with financial resilience may be traced to social capital, one of the components of Salignac et al. (2019)'s Multidimensional Financial Resilience Framework. As financial hardships mounted during the pandemic, it would be reasonable to hypothesize that families and friends would provide financial support to loved ones in need, at the expense of their own financial security. Notably, the association between financial resilience and receiving domestic remittances remained minimal and positive from 2017 to 2021.

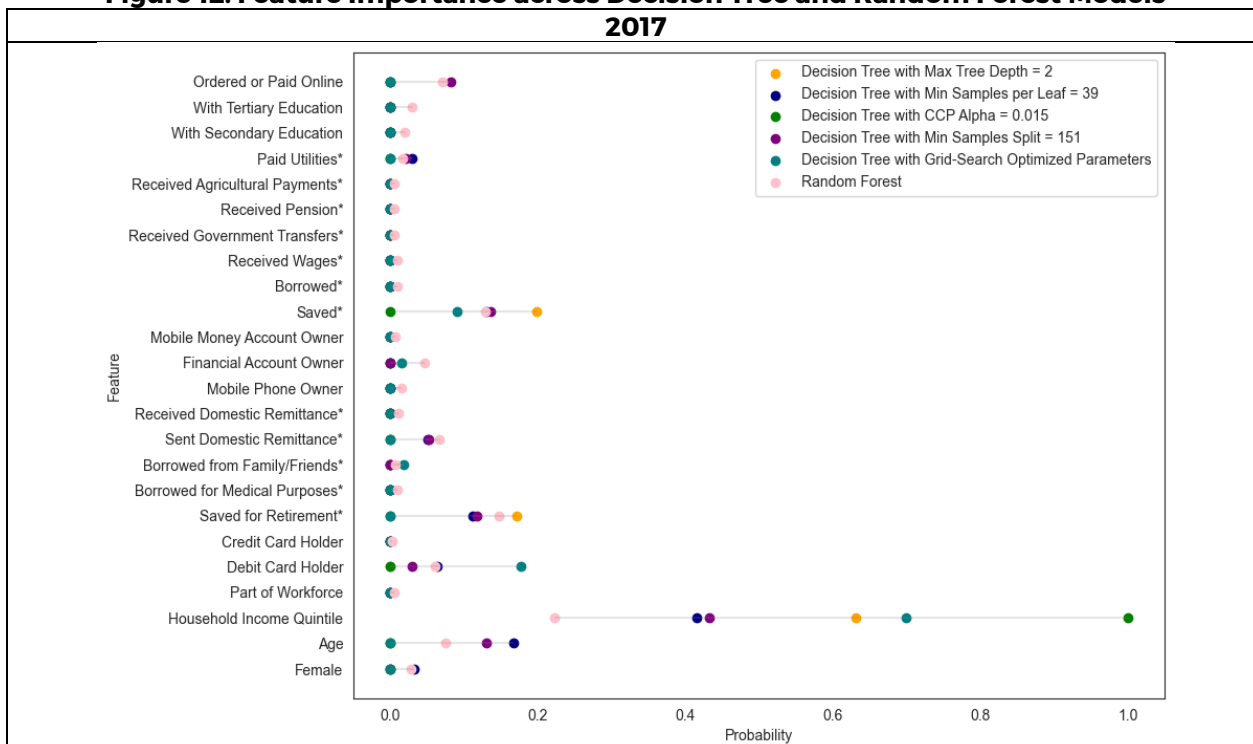
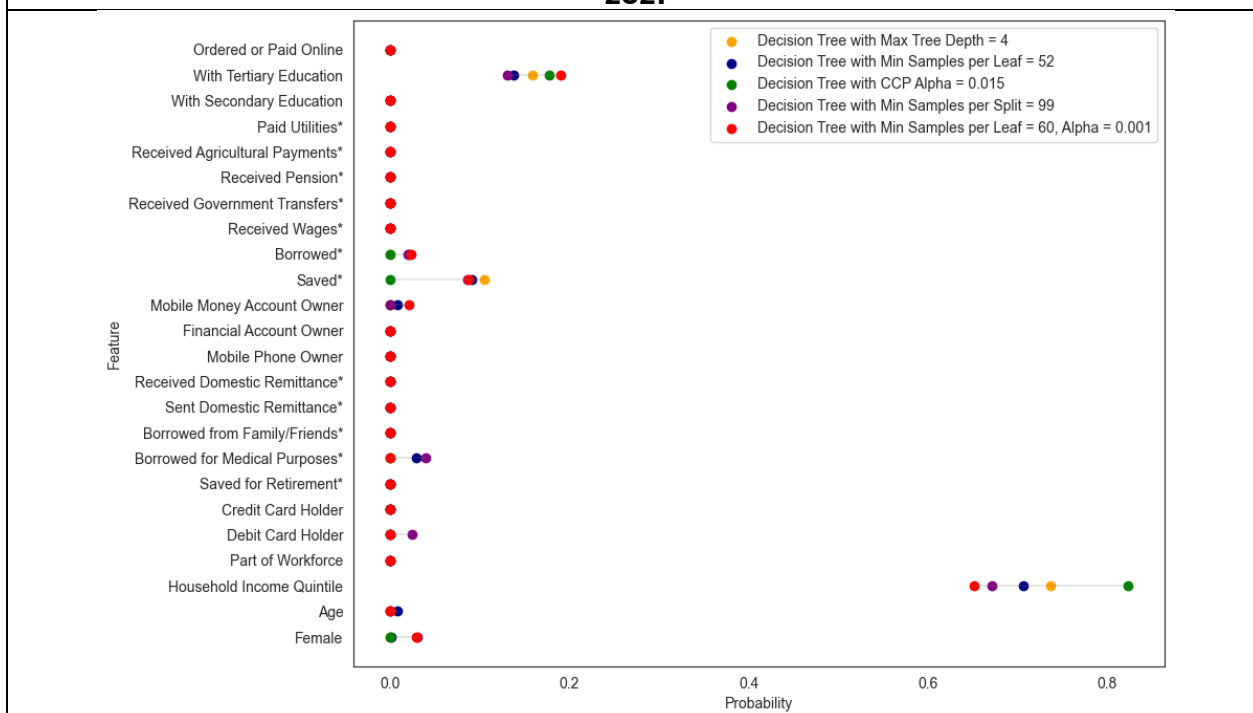
Figure 11 displays the decision trees with the highest true negative rates. Consistent with the results of the logistic LASSO regression, income quintile remained the most important predictor. For the 2017 survey round, this was followed by savings – whether for retirement (for respondents belonging to the richest 40 percent in terms of household income quintile) or regardless of purpose (for the poorest 60 percent). On the other hand, saving behavior did not figure prominently in the 2021 survey round. Instead, the results suggest three groups of respondents exhibiting varying levels of financial resilience: first (and most likely to be resilient) are the more affluent (i.e., among the richest 40 percent) who were also tertiary-educated; second are the rest of the more affluent who did not receive tertiary education; and third (and least likely to be resilient) are respondents belonging to the poorest 60 percent of the population. Nonetheless, it must be noted that Gini scores remain high across most nodes, with a notable exception being tertiary-educated, higher-income respondents for the 2021 survey (who were mostly financially resilient). Respondents must be segregated further, and a larger sample might be needed to identify more accurately who are the financially resilient vis-à-vis those who are not.

**Figure 11: Decision Trees with Highest True Negative Rates
2017 (Minimum Samples per Split = 151)**



Source: Author's estimates

In terms of feature importance for the decision tree and random forest models (Figure 12), the results are similar to the Decision Trees chosen in Figure 11. Income quintile was the top predictor in both the 2017 and 2021 surveys. In 2017, both forms of saving behavior (i.e., saving for any purpose and saving for retirement) also ranked high in feature importance. Other noteworthy factors that figured prominently in 2017 are online purchases, debit card ownership, and age. In 2021, the most predictive factors were tertiary education, saving per se, borrowing for medical reasons, and gender.

Figure 12: Feature Importance across Decision Tree and Random Forest Models
2017**2021**

Source: Author's estimates

VII. Conclusions and Policy Implications

This paper explores the determinants of Filipinos' financial resilience through the construction of predictive models and from there, the generation of inferences on the most important predictors of financial resilience. Using World Bank Global Findex data for the Philippines in 2017 and 2021, empirical results consistently identify income quintile, saving behavior, and gender as the top predictors for financial resilience for both years. Additionally, age, saving for retirement, and online payments were important features for 2017, while tertiary education and medical borrowing were important for 2021. These findings indicate that demographics remains the largest determinant for financial resilience. The results also seem to suggest that financial inclusion, i.e., access to and usage of financial products, does not necessarily translate to resilience.

Income level and gender consistently emerged as the most important variables in the models. Results of the 2021 survey round likewise showed that tertiary education is second only to income quintile in predicting financial resilience. The significance of income, gender, and education to financial resilience is consistent with literature. Debuque-Gonzales and Corpus (2021) point out that lower-income and less-educated Filipinos may experience “involuntarily exclusion” from the formal financial system. In this regard, policymakers may consider crafting programs targeted towards improving employment and educational opportunities for those in the lower socioeconomic classes. Additional safety nets may also be considered for females, who, despite their higher participation in the formal financial system, still lag behind males in terms of financial resilience, with the divide even widening from 2017 to 2021.¹⁰ A similar gender gap is apparent in the labor market, where female labor force participation remains one of the lowest in the region (Buchhave & Belghith, 2022). Initiatives empowering women to gain access to higher quality jobs and encouraging entry into the workforce could thus help close the financial resilience gap.

The Philippine government has implemented a conditional cash transfer (CCT) program which benefits around 20 percent of the population (World Bank, 2017). The decision tree model's choice to split after the 3rd income quintile may indicate that it is not just the poorest of the poor who face increased financial vulnerability. This result warrants a reassessment of social safety nets that have traditionally focused on the poorest 20 percent. Broadening social protection measures provided to the poorest 60 percent may not necessarily involve expanding CCT coverage or providing direct transfers but may come in other forms of assistance such as insurance mechanisms and micro-credit opportunities that may help the entrepreneurial poor to tap into additional funding sources that would allow them to earn income and become self-sufficient. As the 2018 Consumer Finance Survey (CFS) finds, only a small segment of Filipino households (5.1 percent) are engaged in entrepreneurial activities.

Saving behavior was likewise a key predictor in most of the models. Notably, the decision tree identified different forms of savings as important for the different socioeconomic classes. For those with lower income levels, the act of saving, in whatever form or for whatever purpose, mattered the most. However, for those with higher income levels, saving for one's retirement mattered the most.

¹⁰ Females are more likely and may even be more determined to become financially included (Debuque-Gonzales and Corpus, 2021).

The importance for poorer households of having the ability to save, coupled with the emergence of a sizeable association between financial vulnerability and borrowing for medical purposes, highlight the need for enhanced insurance mechanisms against economic shocks such as disasters and medical emergencies. For instance, for lower-income farmers whose earnings may be sensitive to weather-related disruptions, the government may consider setting up agricultural insurance programs. Healthcare coverage may also be reassessed to determine the extent of out-of-pocket payments required from poorer beneficiaries. Improved public health services would likewise improve the quality of life of poor families, protecting them against morbidity-induced income losses and health expenditures. Further, the government should enhance pandemic preparedness, as poor epidemiological response undermines not only public health but also the financial resilience of the lower income population.

For higher-income individuals, saving for the future may matter more than saving for other purposes (e.g., for business or for a major purchase). Since even the more affluent can experience financial vulnerability, expanding financial literacy programs can be beneficial in instilling the value of financial discipline and saving into the broader populace. Both the CFS report and Debuque-Gonzales and Corpus (2021) study similarly advocate for financial education. The 2018 CFS also raised concerns that Filipino households appeared to prioritize consumption and purchase of non-financial assets over savings and investment. More than half of Filipino households were found to be over-indebted and experienced challenges meeting monthly obligations.

Modest model accuracy coupled with other limitations of the dataset, such as the narrow proxy measure for financial resilience used in the Findex survey, may weigh on the validity of the findings. The consistent below-70 percent accuracy of the various models highlights the need for more granular survey data, as several factors remain unmeasured by the current dataset. A more adequate assessment of financial resilience may emanate from the introduction of additional demographic variables relating to home ownership, employment status, industry, as well as information across other dimensions of financial resilience (such as savings level, debt management ability, access to credit, access to insurance, credit demand, insurance demand, financial knowledge, financial behavior, and access to social support).

Findings from this study may allow policymakers to assess priority areas requiring intervention. However, a definitive understanding of the financially vulnerable and the appropriate interventions would require the conduct of more focused studies. As such, future research may consider investigating causal links between financial resilience and the variables with high predictive value that were identified in this study.

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